

Supplementary Material 3: Simulation Tests

Description: *This Supplementary material provides the results concerning the performance of the proposed SLAM algorithm in a simulated arboreal environment.*

The 3D render of the simulated environment, which corresponds to a pine plantation, is shown in Fig. 1 using the rvis visualization tool of ROS. The simulated testing area comprises six parallel tree rows, each 10 *m* in length and spaced at 2.4 *m* intervals. The mobile robot corresponds to a Unitree Go1 legged robot equipped with a Velodyne VLP-16 3D lidar sensor, as illustrated in Fig. 1c. The lidar model assumes a Gaussian zero-mean range error with standard deviation of 0.05 *m* to emulate realistic conditions. The simulation utilizes the 3D scan processing proposed in Section 2.1 of the paper to match the 2D contours of the tree trunks. The grid map resolution for the simulated experiments has a resolution of 0.01 *m/pixel*.

Testing in simulation considers ten realizations of a 2×2 *m* square trajectory and ten realizations of a 6.5×2 *m* rectangular trajectory, assuming no GNSS nor IMU is available. The estimated positions for each run following the reference path (labeled “Ground truth”) are shown in Figs. 2a and 2b. The averaging of the estimated maps is shown in Figs. 2c and 2d. Comparing the estimated trajectories with the actual trajectory and the reconstructed maps with the ground truth map using the metrics provided in the Supplementary Material is possible to validate the effectiveness of the approach, its repeatability, and confirm that the positioning accuracy is limited mainly by the lidar’s measurement noise and the scan matching errors.

Position estimation and reconstructed map quality metrics are summarized in Table 1 and Table 2, respectively. The mean position estimation error and its RMS value are on average below 0.26 *m* for both path, while the angular error is below 0.15 *rad* (8.59°) on average. The position displacement estimation error is smaller than 0.02 *m* and the angular displacement estimation error is smaller 0.06 *rad* (3.44°).

Table 1: Pose estimation errors for the simulated arboreal environment obtained with the proposed approach. Position *p* errors in meters and angular *θ* errors in radians.

Pose estimation metrics	Simulation experiment	
	Path 1	Path 2
$\bar{e}_p \pm C_p$	0.13 ± 0.04	0.23 ± 0.08
$\bar{e}_\theta \pm C_\theta$	0.07 ± 0.02	0.08 ± 0.04
$e_{RMS_p} \pm C_{RMS_p}$	0.15 ± 0.01	0.26 ± 0.02
$e_{RMS_\theta} \pm C_{RMS_\theta}$	0.11 ± 0.00	0.15 ± 0.01
$\bar{e}_{\Delta p} \pm C_{\Delta p}$	0.02 ± 0.01	0.02 ± 0.01
$\bar{e}_{\Delta \theta} \pm C_{\Delta \theta}$	0.03 ± 0.01	0.02 ± 0.03
$e_{RMS_{\Delta p}} \pm C_{RMS_{\Delta p}}$	0.03 ± 0.00	0.03 ± 0.00
$e_{RMS_{\Delta \theta}} \pm C_{RMS_{\Delta \theta}}$	0.03 ± 0.00	0.06 ± 0.01
$e_{p\%} \pm C_{p\%}$	1.00 ± 0.00	1.00 ± 0.00
$e_{end} \pm C_{end}$	0.10 ± 0.00	0.07 ± 0.00

Table 2: Map estimation errors for the simulated arboreal environment obtained with the proposed approach using a grid map resolution of 0.01 *m/pixel*.

Map estimation metrics	Simulation experiment	
	Path 1	Path 2
$Precision \pm C_{Precision}$	0.31 ± 0.05	0.48 ± 0.02
$Sensitivity \pm C_{Sensitivity}$	0.37 ± 0.06	0.70 ± 0.03
$Accuracy \pm C_{Accuracy}$	0.99 ± 0.01	0.99 ± 0.01

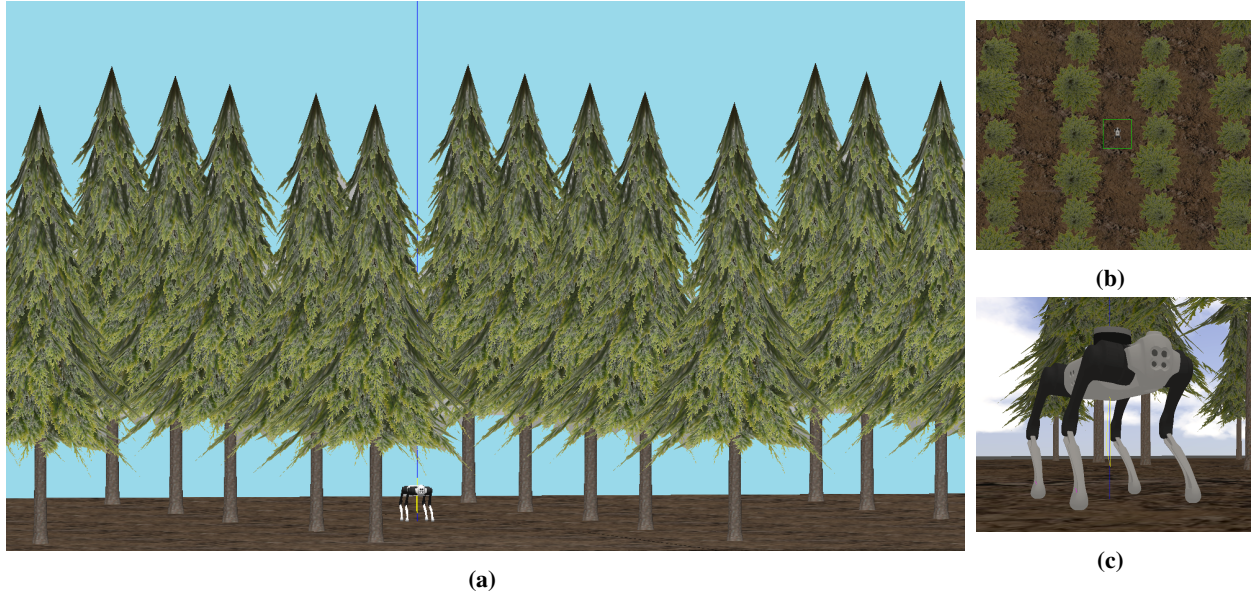


Figure 1: Simulated forest environment: (a) front view of the forest and the mobile robot, (b) top view, (c) robot and lidar close-up.

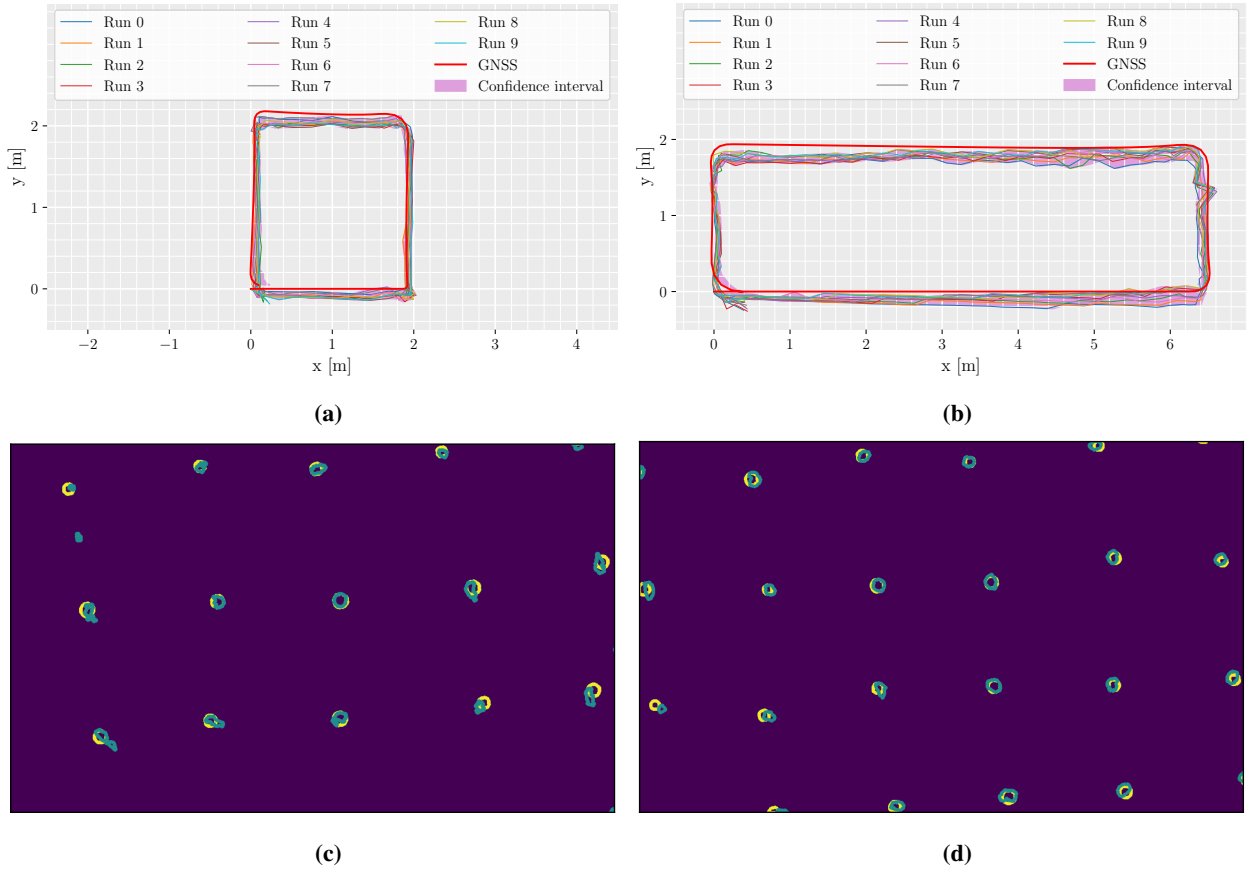


Figure 2: Estimated trajectories and maps using the proposed approach: (a) square path, (b) rectangular path, (c) reconstructed map for the square path, (d) reconstructed map for the rectangular path. Trajectory ground truth is shown in red, and map ground truth is represented as yellow circles (tree trunks). Map area is 12×7 m for the square path and 13×8 m for the rectangular path with 0.01 m/pixel resolution.